

RENTipid ListingBridge v1.0

Complete User, Operations, Developer & Business Reference Manual

Implementation, User Guidance, Operations, Architecture, Security, Deployment and Governance Reference

Controlled Technical & Operational Reference

Module Version: 1.0

Release Tag: listingbridge-v1.0.0-frozen

Application SHA: a8647df71aa9c610027054e2016fd73b53f3b238

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Prepared from frozen release source and evidence ledger. No application, test, Prisma, deployment, production, or feature-flag state is modified by this manual.

Document Control

Table 1. Document control

Field	Value
Document	RENTipid ListingBridge v1.0 Complete Reference Manual
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Intended Audiences

This manual serves four audiences in one coherent document: RENTipid users, RENTipid operators, support, and administrators; developers, engineers, and integrators; owners, management, and product owners.

How To Use This Manual

Part I explains business purpose and implemented scope. Part II is the provider operations and support manual. Part IV is the developer reference. Part V covers integration. Part VI contains appendices, reference tables, and quick cards.

Document Conventions

Table 2. Document conventions

Convention	Meaning
NOTE	Useful clarification.
IMPORTANT	Required behavior or release invariant.
WARNING	Risk that can affect security, data integrity, or release governance.
OPERATOR ACTION	Support or operational action.
DEVELOPER NOTE	Implementation-specific detail.
BUSINESS NOTE	Owner or management decision context.

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Quick Audience Navigation

Table 3. Quick audience navigation

Audience	Start with	Most useful appendices
Provider / user	Chapters 4-11	A, E, J
Operator / support / admin	Chapters 12-17	B, C, E, F, I, J
Developer / maintainer	Chapters 18-34	G, H, D, F
Business owner / product owner	Chapters 1-3 and 35-36	I, J

Part I - Executive & Business Reference

Chapter 1 - Executive Overview

ListingBridge is RENTipid v1.0 functionality for importing an existing provider listing into a native RENTipid draft. It reduces repeated data entry while preserving RENTipid as the system of record. External sources are input channels only. RENTipid remains authoritative for identity, provider ownership, listing state, media storage, policy, publication eligibility, audit, and production records.

The import is intentionally one-time. ListingBridge helps gather and normalize listing details, media candidates, location facts, pricing hints, and provenance, then routes the reviewed result through the existing native draft authority. It does not replace manual listing creation and it does not publish listings.

IMPORTANT
ListingBridge itself does not publish, approve, or make a listing bookable. The output of v1.0 is a native RENTipid Listing with status Draft. Submission, review, approval, and publication remain separate existing RENTipid lifecycle steps.

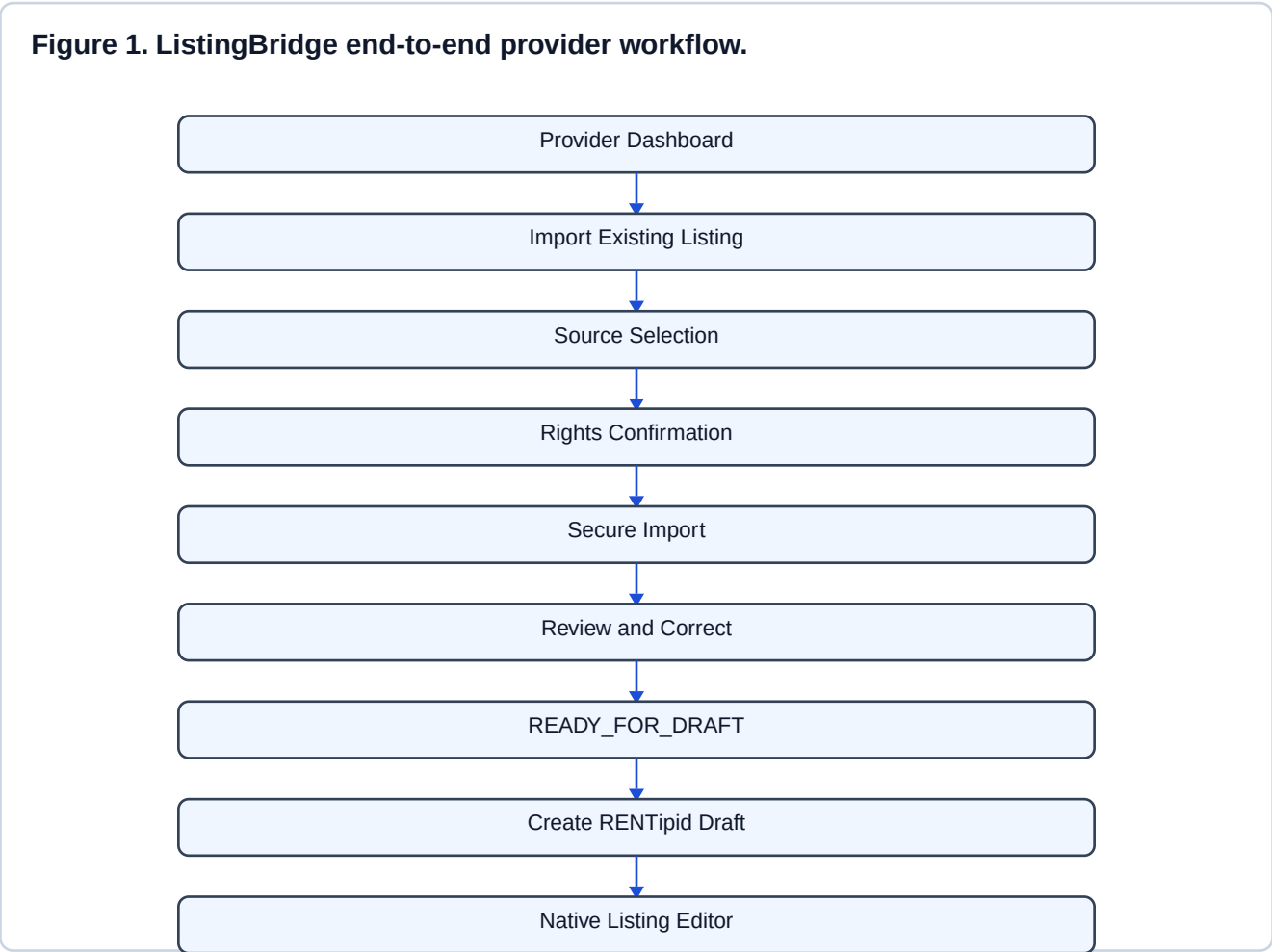


Table 4. v1.0 achievement and non-scope summary

Area	v1.0 achieves	v1.0 intentionally excludes
Onboarding	Provider can import into a reviewable draft workflow.	Continuous OTA synchronization.
External sources	Connector architecture, structured file capability, guarded URL capability, internal test,connector for non-production validation.	Live Airbnb, Agoda, Booking.com, reservation write-back, or broad price write-back.
Controlled Technical & Operational Reference		1.0

Area	v1.0 achieves	v1.0 intentionally excludes
Governance	Frozen release with G1-G13 evidence.	Informal post-freeze modification.
Safety	Server authorization, SSRF controls, provenance, duplicate checks, and audit.	Security bypasses or client-only draft authority.

Chapter 2 - Business Capability & Value

The business capability is controlled external-data ingestion for providers. It improves onboarding efficiency by allowing providers to begin from known listing facts, while still requiring provider review and RENTipid policy controls before any draft enters the normal listing lifecycle.

Table 5. Business benefits table

Capability	Business Value	Risk Control
One-time import to draft	Shortens provider onboarding and reduces duplicate data entry.	Provider review and draft-only boundary prevent uncontrolled publication.
Canonical contract	Creates reusable foundation for future source channels.	External source objects never write directly to Listing.
Field confidence and provenance	Improves support visibility and user trust.	Ambiguous or missing data is surfaced rather than fabricated.
Duplicate detection	Protects marketplace quality.	Exact matches block readiness; possible matches require review.
Feature flags	Allows controlled rollout by capability.	Master kill switch disables import without harming manual listings.
Security-first retrieval	Enables controlled external input.	SSRF, MIME, size, redirect, and rate controls fail closed.

BUSINESS NOTE

The connector framework creates future extensibility, but v1.0 does not include active commercial OTA integrations or bidirectional channel management.

Future connector potential should prioritize authorized API/OAuth or approved partner integrations, followed by PMS/channel-manager feeds and structured files. Unrestricted scraping should not be used to bypass access controls, terms, security review, or provider authorization.

Chapter 3 - Implemented Scope & Limitations

ListingBridge v1.0 implements the import foundation, canonical data contract, durable job model, provider review, corrections, media/location/duplicate intelligence, AI-assisted mapping boundary, observability, and native draft creation integration. The implementation is additive and frozen at application SHA a8647df71aa9c610027054e2016fd73b53f3b238.

Table 6. Source mode hierarchy and frozen availability

Tier	Mode	Architecture support	v1.0 production posture
Tier 1	Authorized API/OAuth	Connector interface supports server-side API/OAuth authorization.	No live partner connector; API_CONNECTORS=false.
Tier 2	PMS/channel manager	Tier is modeled for managed partner feeds.	Future scope.
Tier 3	Structured file	Structured file import capability is implemented in architecture and tests.	FILE_IMPORT=true, but GLOBAL=false blocks production rollout.
Tier 4	Permitted public URL	Secure URL retrieval capability is implemented with SSRF protection.	URL_IMPORT=false.
Tier 5	Manual setup	Fallback to native manual listing wizard.	Available independently of ListingBridge.

WARNING

Do not imply Airbnb, Agoda, or Booking.com are live integrations. They are future roadmap only and are not included in ListingBridge v1.0 frozen release.

The Internal Test Connector is deterministic and intended for development, automated tests, Preview, and controlled OAT validation only. Its descriptor is approved for LOCAL, TEST, and PREVIEW, but has PRODUCTION: DISABLED and INTERNAL_ONLY feature status.

Part II - Provider / End-User Manual

Chapter 4 - User Roles & Prerequisites

A provider must be authenticated and must have authority to create listings for the property or rental item being imported. ListingBridge is not a substitute for ownership, management rights, KYC, policy review, or publication approval.

- Use a verified provider account where required by RENTipid policy.
- Confirm you own or manage the property, are authorized to submit the information, can reuse submitted media, and accept accuracy responsibility.
- Understand that imported information becomes a draft only after review and server-side readiness validation.
- Continue through the normal Listing Editor and separate submission flow when the draft is ready.

Chapter 5 - Getting Started

Providers enter ListingBridge from the provider listing workflow. The implemented route is /dashboard/provider/listings/import. Manual listing creation remains available at /dashboard/provider/listings/new.

- Open Provider Dashboard.
- Choose Import Existing Listing.
- Select an available source. In Production v1.0, ListingBridge global rollout is disabled; controlled Preview/OAT used the Internal Test Connector.
- Continue to authorization.
- Confirm listing authority, media rights, and accuracy responsibility.
- Begin secure import.
- If the source is unavailable or unsuitable, use Create New Listing Directly to open the standard manual wizard.

Chapter 6 - Complete ListingBridge User Workflow

The user workflow is intentionally review-first. Imported source values are not silently published and should be treated as a starting point for provider confirmation.

Table 7. Provider workflow procedure

Step	Action	Result
1	Open Provider Dashboard.	Listing actions are shown.
2	Choose Import Existing Listing.	ListingBridge import page opens.
3	Select available source.	Source card is selected.
4	Continue to authorization.	Rights confirmation stage opens.
5	Confirm authority and media rights.	Required legal/accuracy attestation is captured.
6	Begin secure import.	Server action creates a durable import job.
7	Import, extraction, normalization.	Canonical contract and field records are prepared.
8	Review imported information.	Fields display values and confidence states.
9	Understand confidence states.	Provider can distinguish verified, review, conflict, missing, and prohibited values.
10	Review provenance.	Source hashes and field records preserve traceability.
11	Correct a field.	Save Correction persists a provider resolution.
12	Resolve conflicts/missing info.	Blocking reasons are removed when valid.
13	Confirm readiness.	Server readiness requires no blockers.
14	Proceed to draft.	READY_FOR_DRAFT screen appears.
15	Create RENTipid Draft.	Native ListingService.createDraft is invoked server-side.
16	Open draft in Listing Editor.	Real Listing.id route opens.
17	Continue normal workflow.	Provider edits photos, pricing, terms, and details as needed.
18	Submit for review separately.	Publication is not automatic.

Chapter 7 - Understanding Field Status & Confidence

Table 8. Field status and confidence reference

State	Meaning	Provider action	Readiness impact
VERIFIED	Confirmed by provider or deterministic trusted validation.	Review for accuracy; no action unless wrong.	Normally non-blocking.
HIGH_CONFIDENCE	Imported and normalized with strong evidence.	Review and edit if needed.	Normally non-blocking.
REVIEW_RECOMMENDED	Usable but uncertain or AI/system assisted.	Inspect and confirm or correct.	May warn; can block if required by context.
CONFLICT	Contradictory, invalid, or inconsistent data was detected.	Choose the correct value or correct the field.	Blocks required fields and critical checks.
MISSING	Required or useful information was not found.	Enter the missing information manually.	Blocks when required or readiness-critical.
PROHIBITED	Source data is not allowed into listing content.	Do not reuse; provide allowed replacement content if needed.	Active prohibited content blocks.

Chapter 8 - Reviewing & Correcting Imported Data

The review screen distinguishes imported values from provider-modified values. Editable fields allow a provider to open an edit dialog, enter a correction, and save the correction. The server action validates ownership against the persisted import job before saving the resolution.

A saved correction creates or updates ListingImportResolution with resolution_type PROVIDER_OVERRIDE and updates ListingImportField with the corrected normalized value, VERIFIED confidence, provider_modified=true, and validation_state=VALIDATED. This produces a durable correction trail tied to the authenticated provider.

NOTE

Prohibited fields cannot be accepted into listing content. Providers must supply allowed replacement text or omit that content.

Chapter 9 - Photos, Media & Location

Imported media candidates are supplemental. ListingBridge validates media before storage using byte-level MIME sniffing, size limits, and SHA-256 hashing. Duplicates within the same job reuse the existing stored asset path and are marked `SKIPPED_DUPLICATE`. Invalid media is rejected without necessarily failing the whole import.

Location data is normalized through the RENTipid address normalizer and checked against Philippine bounds when coordinates are provided. Coordinates outside 4.5 to 21.5 latitude or 116.0 to 127.0 longitude conflict with Philippine listings and can block readiness. Providers must review uncertain or conflicting location data.

Chapter 10 - Duplicate Listings

Duplicate checks protect providers and the marketplace from repeated drafts for the same property. The implemented detector considers same source reference, same provider and address/city with title similarity, coordinate proximity within 50 meters, and similar title in the same city for different providers.

Table 9. Duplicate match levels

Level	Threshold / signal basis	Effect
EXACT_MATCH	Max signal score >= 0.95, including same source reference or near-coordinate same-provider match.	Blocks draft readiness.
LIKELY_MATCH	Max signal score >= 0.75.	Requires review.
POSSIBLE_MATCH	Max signal score >= 0.50.	Requires review.
NO_MATCH	No meaningful duplicate signals.	No duplicate blocker.

Repeat Create Draft does not create duplicate
draft service returns the existing Listing.id if

Chapter 11 - User Troubleshooting

Table 10. User troubleshooting table

Symptom	Possible cause	User action	Escalate when
Source unavailable	Connector disabled or feature flag off.	Use manual listing creation or retry later.	Expected source should be enabled for your account.
Authorization rejected	Rights confirmation missing or connector authorization failed.	Confirm authority and media rights; reauthorize if prompted.	You believe authorization is valid but still blocked.
Imported field missing	External source did not expose the field.	Enter the value manually during review.	The field repeatedly disappears after saving.
Field says Please Review	Confidence is review-recommended or AI-assisted.	Inspect and confirm or edit the value.	The status does not change after correction.
Conflict shown	Location, duplicate, or field validation conflict.	Resolve the highlighted field.	You cannot determine the correct value.
Unsafe URL rejection	URL failed SSRF, protocol, DNS, redirect, or size policy.	Use another approved source or manual listing.	Never attempt to bypass URL safety.
Media failure	Invalid MIME, too large, duplicate, retrieval failure, or storage issue.	Upload photos through the normal editor if needed.	Many unrelated imports fail media processing.
Duplicate detected	Same source, nearby coordinates, or similar title/location.	Open existing draft/listing or clarify unique details.	System marks a clearly different property as exact match.
Cannot reach draft readiness	Missing required field, unconfirmed rights, blocking duplicate, invalid media, or location conflict.	Complete each listed blocker.	Blockers persist after correction.
Import retry	Transient network or upstream issue.	Retry if offered; otherwise use manual fallback.	Retry fails repeatedly.
Session/authentication issue	Login expired or role not authorized.	Sign in again with provider account.	You are verified but still denied.

Part III - Operations & Support Manual

Chapter 12 - Operating Model

The frozen production posture is safe-by-default: LISTINGBRIDGE_GLOBAL=false. Operators should treat ListingBridge as a controlled import subsystem that can be disabled without affecting native manual listing creation or existing listings.

- Monitor feature flag posture, connector health, import job state, retries, security blocks, media failures, draft creation failures, and audit events.
- Support boundaries: guide providers through review and manual fallback; do not fabricate missing facts or bypass security.
- Expected roles include support operators, administrators, SRE/on-call, security operations, release manager, and engineering maintainer.

Chapter 13 - Feature Flags

Frozen Production feature flags are recorded in the freeze manifest. LISTINGBRIDGE_GLOBAL=false is the master kill-switch and safe rollout state. It blocks connector evaluation and the import entry point while preserving manual listing creation.

Table 11. Feature flag reference

Key	Purpose	Frozen Production Value	Effect when OFF	Effect when ON	Dependencies	Operational caution
LISTINGBRIDGE_GLOBAL	Master ListingBridge kill switch and rollout gate.	false	No ListingBridge connectors are offered; manual listing creation remains available.	ListingBridge import entry points and connector evaluation are permitted subject to child flags.	All ListingBridge capabilities.	Frozen release flag. Change only through controlled rollout.
LISTINGBRIDGE_FILE_IMPORT	Enables Tier 3 structured file import capability.	true	Structured file import is unavailable.	Structured file connectors can be evaluated when global flag also permits.	LISTINGBRIDGE_GLOBAL.	Do not confuse enabled capability with production-wide public rollout.
LISTINGBRIDGE_URL_IMPORT	Enables Tier 4 public URL retrieval.	false	URL import is unavailable.	SSRF-guarded URL retrieval can run for approved connectors.	LISTINGBRIDGE_GLOBAL, retrieval policy, SSRF controls.	High caution due to external network exposure.
LISTINGBRIDGE_API_CONNECTORS	Enables API or OAuth partner connectors.	false	OAuth/API connector paths are unavailable.	Approved server-side API connectors may be evaluated.	LISTINGBRIDGE_GLOBAL, connector credentials, compliance approval.	No live Airbnb, Agoda, or Booking.com connector exists in v1.0.
LISTINGBRIDGE_MEDIA_IMPORT	Enables media retrieval, validation, hashing, and storage.	true	Imports can continue as text/detail-only where supported.	External media candidates may be ingested after validation.	Storage provider, media MIME validation, source retrieval.	Disable if storage or MIME validation anomaly is observed.
LISTINGBRIDGE_AI_MAPPING	Enables bounded semantic mapping assistance.	true	Deterministic mapping and provider review continue without AI.	AI may suggest amenities, categories, summaries, and original descriptions from verified facts.	Unified AI adapter and read-only/draft-only tools.	AI output is advisory and never policy-authoritative.
LISTINGBRIDGE_AVAILABILITY_IMPORT	Enables external availability/calendar import.	false	Availability data is not imported from external calendars.	Approved availability connector capability may be used.	LISTINGBRIDGE_GLOBAL and connector availability support.	Reservation write-back and continuous sync are not v1.0 scope.

Chapter 14 - Job Lifecycle & Support Diagnostics

Table 12. Durable state transition reference

State	Meaning	Expected transition	Operator concern	Recovery / escalation notes
CREATED	Durable job record exists.	AUTHORIZING or FETCHING.	Job should not remain idle for long.	Retry or cancel only through controlled support flow.
AUTHORIZING	Provider rights or connector authorization is being verified.	FETCHING.	Repeated authorization failures may indicate user action or connector issue.	Ask provider to retry authorization; do not bypass rights confirmation.
FETCHING	Connector or secure retrieval is obtaining source payload.	EXTRACTING.	Timeouts and connector errors are expected transient concerns.	Retry if retryable; security failures become final.
EXTRACTING	Structured facts are being read from payload.	NORMALIZING.	Malformed payloads can produce missing fields.	Escalate only if repeated for known-good source.
NORMALIZING	Facts are mapped into canonical contract.	PROCESSING_MEDIA or VALIDATING.	Unexpected confidence spikes or prohibited fields require review.	Inspect normalized field records and rejected fields.
PROCESSING_MEDIA	Media candidates are downloaded, validated, deduped, and stored.	VALIDATING, NEEDS_REVIEW, or READY_FOR_DRAFT.	Storage, MIME, or partial failures.	Text import may survive partial media failure; missing required photo can block readiness.
VALIDATING	Readiness, policy, location, duplicate, and field checks run.	NEEDS_REVIEW or READY_FOR_DRAFT.	Blocking fields must be resolved before draft.	Use field/resolution records to guide provider.
NEEDS_REVIEW	Provider must inspect, correct, or confirm fields.	READY_FOR_DRAFT or VALIDATING.	Support should not edit provider assertions casually.	Provider correction is preferred.
READY_FOR_DRAFT	Server-side readiness has no blockers.	CREATING_DRAFT.	If job regresses, review duplicate/location/media blockers.	Proceed to draft only after rights confirmation.
CREATING_DRAFT	Draft creation through native Listing authority is underway.	COMPLETED or failure state.	Stuck jobs may have native draft already created.	Check created_listing_id before any retry.
COMPLETED	Native RENTipid draft has been created and linked.	Terminal.	No further import mutation expected.	Provider continues normal listing editing workflow.
FAILED_RETRYABLE	Transient failure occurred and may be retried.	Prior processing state, FAILED_FINAL, or CANCELLED.	Retry count and next_attempt_at matter.	Retry is bounded by max_retries=3.
FAILED_FINAL	Non-retryable or exhausted failure.	Terminal.	Security, corrupt data, or repeated retry exhaustion.	Manual listing fallback; preserve audit evidence.
CANCELLED	Provider or operator cancelled import.	Terminal.	No draft should be created from a cancelled job.	Manual fallback remains available.

Figure 2. Durable import job lifecycle.



Chapter 15 - Monitoring & Observability

Implemented observability includes a ListingBridgeHealthDiagnosticsService, connector health snapshots, durable job timestamps, retry fields, structured logs, alert evaluation, and a metrics collector with safe dimensions. The production health endpoint is /api/health and production verification recorded HTTP 200 with database connected.

Table 13. Implemented metric and signal reference

Metric / signal	Purpose
listingbridge_import_started_total	Counts import starts.
listingbridge_import_completed_total	Counts completed imports.
listingbridge_import_failed_total	Counts failed imports.
listingbridge_connector_failure_total	Tracks connector-level failures.
listingbridge_ssrf_block_total	Tracks blocked SSRF/security retrieval attempts.
listingbridge_rate_limit_total	Tracks retrieval rate limiting.
listingbridge_media_failure_total	Tracks media processing failures.
listingbridge_duplicate_detected_total	Tracks duplicate detection results.
listingbridge_review_required_total	Tracks jobs requiring provider review.
listingbridge_ai_fallback_total	Tracks AI-disabled or fail-closed fallback.
listingbridge_draft_created_total	Counts native draft creations.
listingbridge_draft_creation_failure_total	Counts draft creation failures.

IMPORTANT

Metrics use bounded dimensions only: environment, connectorId, resultClass, stage, failureCategory, and aiEnabled. Raw URLs, user IDs, titles, tokens, and customer PII must not be dimensions.

Chapter 16 - Incident Response

Table 14. Incident response escalation matrix

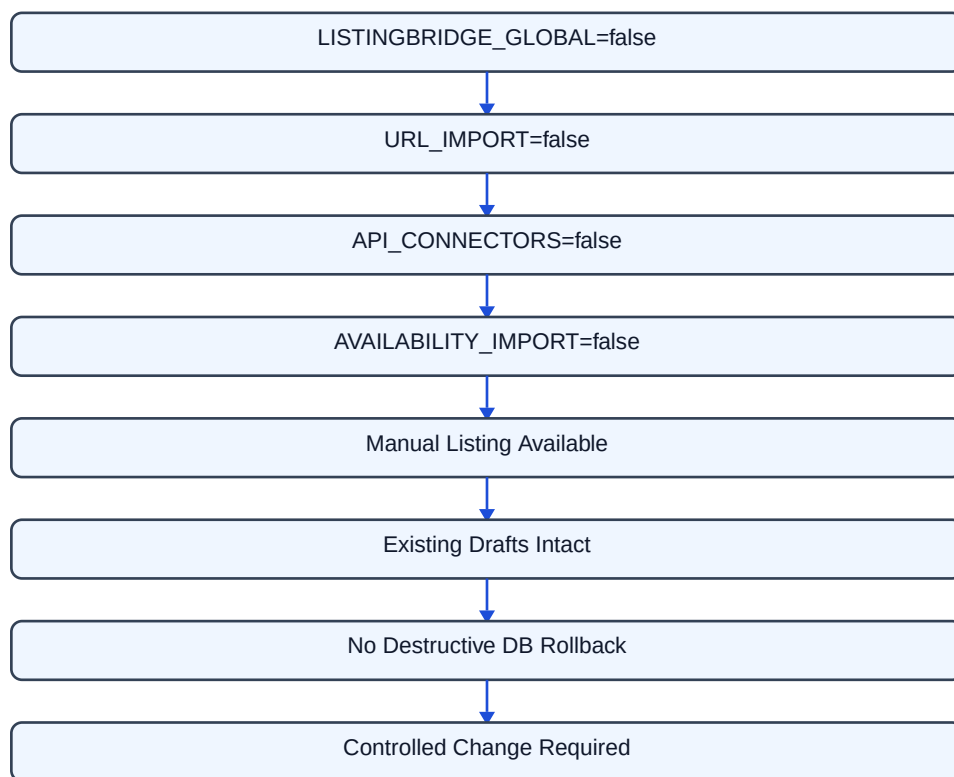
Incident	Immediate action	Escalate to
Connector outage	Check connector health, upstream status, and flags; keep manual fallback available.	SRE and connector owner.
External-source failure	Classify retryable vs final; preserve user-facing message.	Engineering if reproducible.
Repeated import failure	Review status, last_error_code, retry_count, and source mode.	Engineering/SRE.
SSRF/security block spike	Treat as security event; review actor/source patterns; rate limit or suspend if abusive.	SOC/Security.
Worker interruption	Resume from durable state; never create a draft without checking created_listing_id.	Engineering.
Media processing failure	Check storage and MIME failures; consider MEDIA_IMPORT=false for systemic issue.	SRE/Storage owner.
Duplicate anomalies	Inspect duplicate signals and thresholds.	Product and Engineering.
Draft creation failure	Check readiness blockers, category resolution, and native ListingService constraints.	Engineering.
Database issue	Use standard DB incident process; avoid destructive schema rollback.	SRE/DBA.
Runtime regression	Set GLOBAL=false; consider Vercel rollback to known-good release.	Release manager.

Chapter 17 - Rollback & Kill Switch

Feature rollback for the frozen release is LISTINGBRIDGE_GLOBAL=false. Subsystem rollback flags include LISTINGBRIDGE_URL_IMPORT=false, LISTINGBRIDGE_API_CONNECTORS=false, and LISTINGBRIDGE_AVAILABILITY_IMPORT=false.

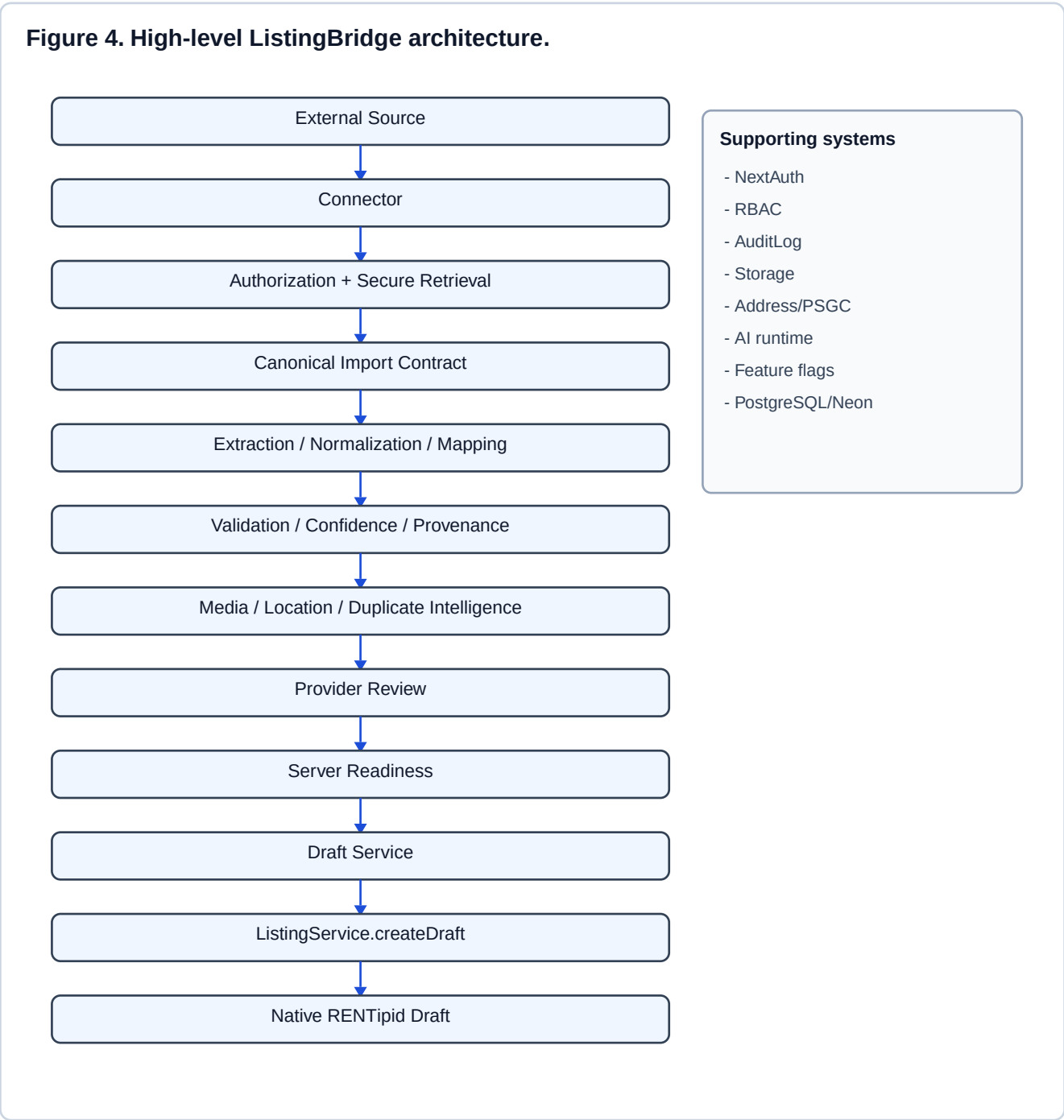
- Manual listing creation remains available and independent.
- Existing listings and drafts remain intact.
- No destructive schema rollback is expected because the ListingBridge migration is additive.
- Use Vercel application rollback where appropriate for runtime regressions.
- Use forward-fix for additive DB migration issues through reviewed change control.
- Never run unreviewed production commands or expose secrets in support notes.

Figure 3. Feature-flag and rollback model.



Part IV - Developer / Engineering Reference

Chapter 18 - Architecture Overview



ListingBridge is implemented as an additive provider UI under `src/components/listings/listings/import/actions.ts`. The native draft au ListingService.createDraft.

Chapter 19 - Reused Authorities

Table 15. Verified reused authorities

Authority	Repository path	ListingBridge use
Authentication	src/lib/auth.ts; src/app/api/auth/[...nextauth]/route.ts	NextAuth v4 server session validation.
Provider identity	prisma/schema.prisma: User, UserProfile, BusinessProfile	Associates import jobs and drafts to provider_id.
RBAC	src/lib/permissions.ts; src/lib/security/permissions.ts	Provider/admin access enforcement.
Draft authority	apps/api/src/services/listingService.ts	ListingService.createDraft creates native Draft listing.
Publication authority	Existing Listing.status and listing service lifecycle	Publication remains outside ListingBridge.
Policy	src/lib/listingbridge/normalization/prohibited-filter.ts and existing prohibited item authorities	Prohibited source data is filtered and marked.
Location	src/lib/address/AddressService.ts; src/lib/address/normalizer.ts	Address normalization and Philippine location handling.
Storage	src/lib/listingbridge/media/media-storage.ts; src/lib/security/upload-security.ts	Validated media storage and upload-security reuse.
AI	src/lib/listingbridge/ai/*; src/lib/ai/tools/AIToolGateway.ts	Read-only and draft-only bounded tools.
Audit	src/lib/audit.ts; ListingImportAuditEvent	Lifecycle and security event trail.
Feature flags	SystemSetting; src/lib/listingbridge/connectors/feature-flags.ts	Runtime capability gates.
Database	PostgreSQL/Neon/Prisma	Durable import records and native Listing records.

Chapter 20 - Connector Architecture

The canonical connector contract is ListingBridgeConnector in src/lib/listingbridge/connectors/types.ts. It declares config, identifySource(), getCapabilities(), authorize(), fetchListing(), fetchMedia(), fetchAvailability(), normalize(), validateResponse(), and healthCheck().

Figure 5. Connector architecture and control boundary.

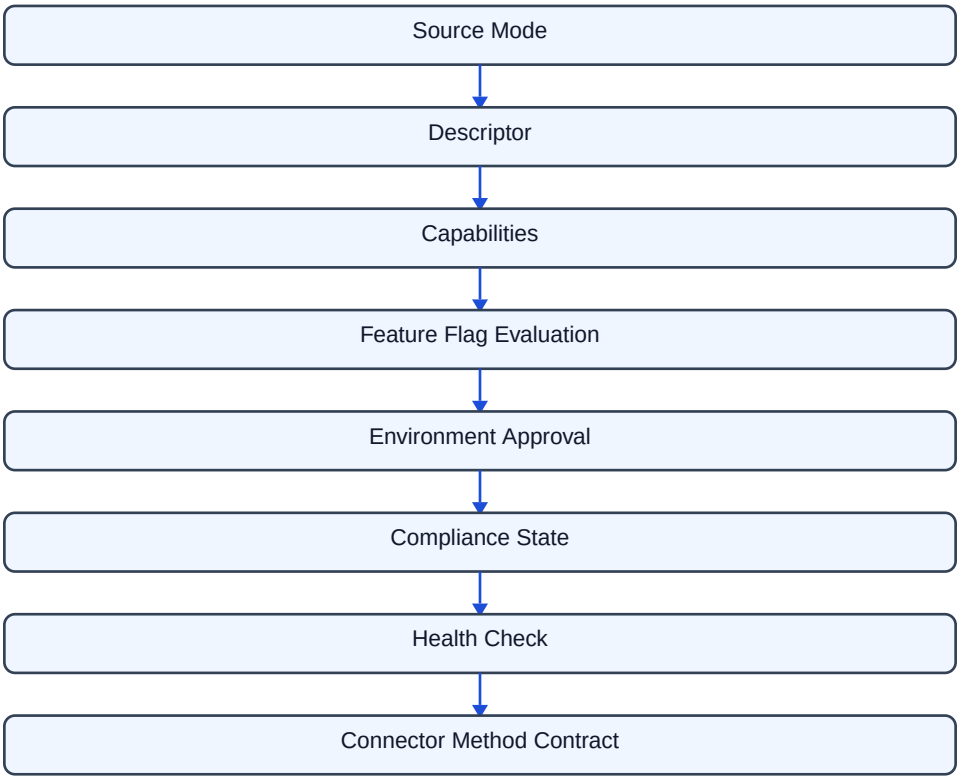


Table 16. Connector control fields

Control	Purpose
Capability declaration	Advertises listing facts, media, availability, structured file, URL retrieval, API/OAuth, rights confirmation, and AI-assisted mapping support.
Environment status	LOCAL, TEST, PREVIEW, and PRODUCTION each declare APPROVED/DISABLED/REVIEW_REQUIRED/BLOCKED.
Feature flags	Required global and capability flags are evaluated before availability.
Authorization type	NONE, rights confirmation, server-side API key/OAuth, signed URL, file upload, public URL, or manual input.
Retry and timeout policy	Per-connector bounded attempts, delays, redirects, and response size.
Compliance and health	Compliance must be approved and health cannot be disabled/unhealthy for availability.

DEVELOPER NOTE

The Internal Test Connector id is internal.test.fixture. It is deterministic, version 1.0.0, TIER_3_FILE, INTERNAL_ONLY, PREVIEW-approved, and Production-disabled.

Chapter 21 - Canonical Import Contract

The canonical import contract lives in `src/lib/listingbridge/types/canonical-contract.ts` with schemaVersion `rentipid.listingbridge.v1`. It is parsed through a Zod schema before persistence or downstream use.

Table 17. Canonical contract groups

Group	Purpose
source	Connector id, tier, source reference hash/label, authorization method, extraction timestamp.
identity	Provider id, optional import job id, idempotency key.
property	Title, description, category suggestion, condition, property type.
location	Raw address, city, province, country, postal code, latitude, longitude, PSGC code.
capacity	Quantity, max guests, bedrooms, bathrooms.
rooms	Room names, types, bed counts, sleeps values.
amenities[]	Canonical amenity terms.
rules	General rules, duration limits, pickup/delivery flags, delivery fee.
pricingHints	Hourly/daily/weekly/monthly rates, deposit, replacement value, PHP currency.
availability	Availability dates, source calendar hash, provider confirmation flag.
media[]	Source reference hash, label, caption, cover flag, order, MIME, content hash, confidence.
provenance	Raw payload hash, AI assisted flag, AI non-authority, model version, fact count, corrections, rejected fields.
fieldConfidence	Per-field confidence state, score, authority, provenance, review and confirmation markers.
unresolvedFields[]	Blocking or optional gaps with expected correction source.

External source objects never write directly to Listing. They are normalized into this contract, validated, then mapped to the native draft payload only after readiness.

Chapter 22 - Data Mapping, Provenance & Confidence

The normalization pipeline uses deterministic aliases in StructuredFactExtractor, taxonomy mapping for property and amenities, prohibited data filtering, commercial/rule classification, conflict detection, optional bounded AI suggestions, and Zod validation. Source values are hashed and attached to provenance rather than exposing sensitive source data.

Provider corrections become ListingImportResolution records and update field records with provider_modified=true. AI-assisted values remain advisory and reviewable; contract validation explicitly marks aiOutputAuthoritative=false.

Chapter 23 - Database Model

Figure 6. Simplified ListingBridge database relationship diagram.

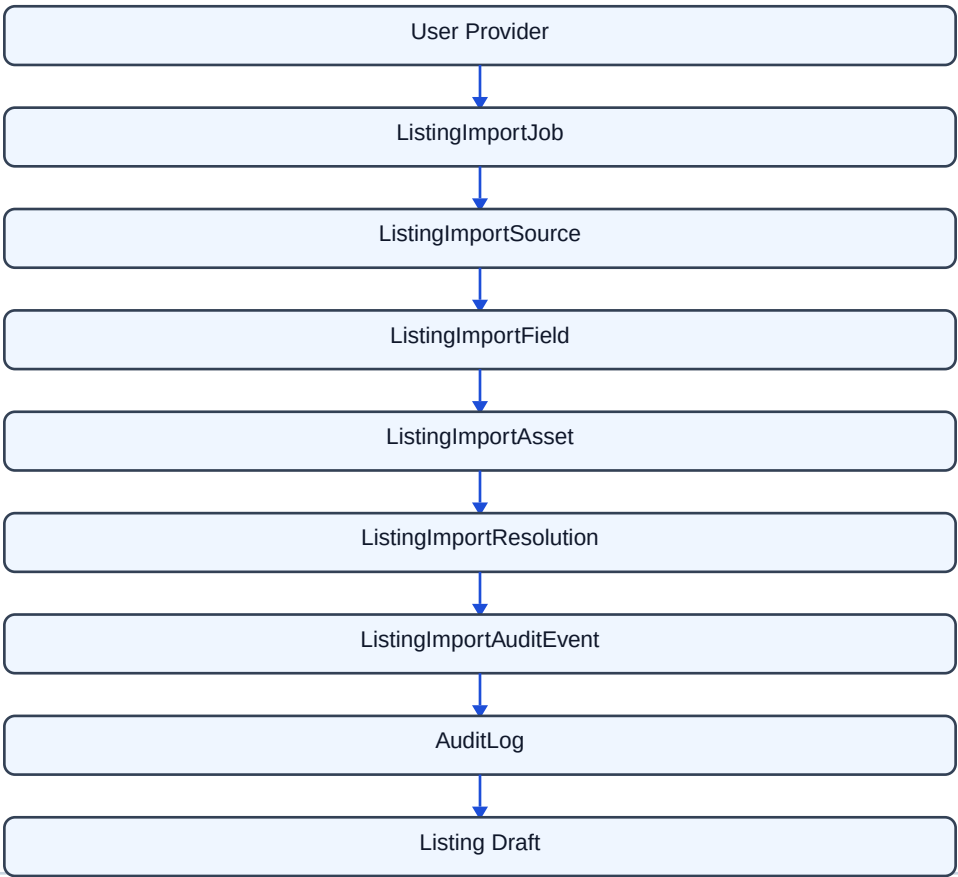


Table 18. ListingBridge additive entities

Entity	Purpose	Key relationships	Lifecycle role	Important constraints
ListingImportJob	Durable root import job and status state.	provider_id -> User; created_listing_id -> Listing.	Owns canonical payload, retry metadata, status, and draft linkage.	Unique idempotency_key and created_listing_id; indexes by provider/status/source.
ListingImportSource	Per-source retrieval/provenance record.	job_id -> ListingImportJob.	Records connector, tier, source mode, version, hashes, and retrieval metadata.	Indexes source reference and identifier.
ListingImportField	Field-level normalized value, confidence, and validation record.	job_id -> ListingImportJob; optional source_id -> ListingImportSource.	Supports provider review, readiness, and provenance.	Unique job_id+field_name; indexes confidence and blockers.
ListingImportAsset	Media candidate, validation, storage, and hash record.	job_id -> ListingImportJob.	Tracks media success, rejection, duplicates, and storage path.	Unique job_id+source_reference_hash and job_id+content_sha256.
ListingImportResolution	Provider/system resolution of a field.	job_id -> ListingImportJob; resolved_by_user_id -> User.	Persists corrections and rights confirmation.	Unique job_id+field_name; indexed by resolver/time.
ListingImportAuditEvent	Append-only import audit trail.	job_id -> ListingImportJob; actor_user_id -> User; audit_log_id -> AuditLog.	Records lifecycle, authorization, security, correction, and draft events.	Indexed by job/time, actor/time, and central audit link.
Controlled Technical & Operational Reference		29		1.0

Chapter 24 - Migrations & Production Database History

The canonical release migration is 20260831000000_add_listingbridge_import_job_foundation. It is additive: it creates four enums and six new ListingBridge tables plus indexes and foreign keys to User, Listing, and AuditLog.

Frozen production migration count is 60. During the controlled G9 deployment, 59 historical production migrations were baselined before the ListingBridge migration. This reconciliation was required so Prisma migrate status could become clean without replaying already-applied historical production changes.

WARNING

This history is not an instruction for casual manual database manipulation. Future schema changes require reviewed migration/change-control procedures and appropriate environment isolation.

Chapter 25 - Import Repository & Durable Jobs

ListingImportRepository creates or returns jobs by idempotency key, attaches source records, upserts field provenance, validates state transitions, stores canonical payloads, increments retry count for retryable failures, and records status-change audit events. Durable state includes provider_id, source connector/tier, hashes, status, retries, lock metadata, errors, timestamps, and optional created_listing_id.

Chapter 26 - Readiness & Draft Creation

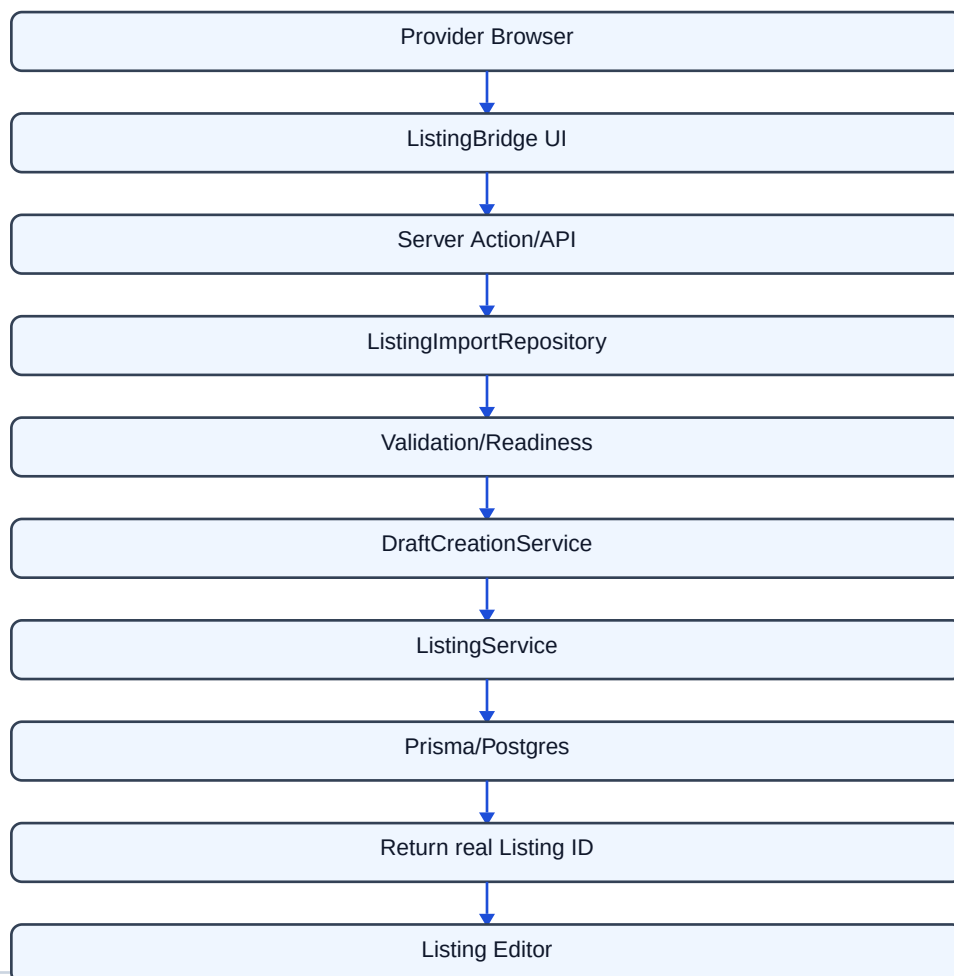
Final implementation corrected the browser flow exposed during OAT. Earlier OAT found a client-only mock draft id that caused a draft editor 404. The frozen implementation uses Server Actions and real database persistence.

The browser can initiate the action and submit the current review snapshot, but it cannot create a native draft by itself. `createNativeDraftAction` re-resolves the authenticated session, fetches the persisted import job, verifies `job.provider_id` equals `session.user.id`, checks idempotent reuse of `created_listing_id`, then delegates to `ListingBridgeDraftCreationService`. The service recalculates readiness, maps the reviewed canonical snapshot to a native payload, invokes `ListingService.createDraft`, records `created_listing_id`, writes `DRAFT_COMMITTED` audit evidence, and returns the actual `Listing.id`.

IMPORTANT

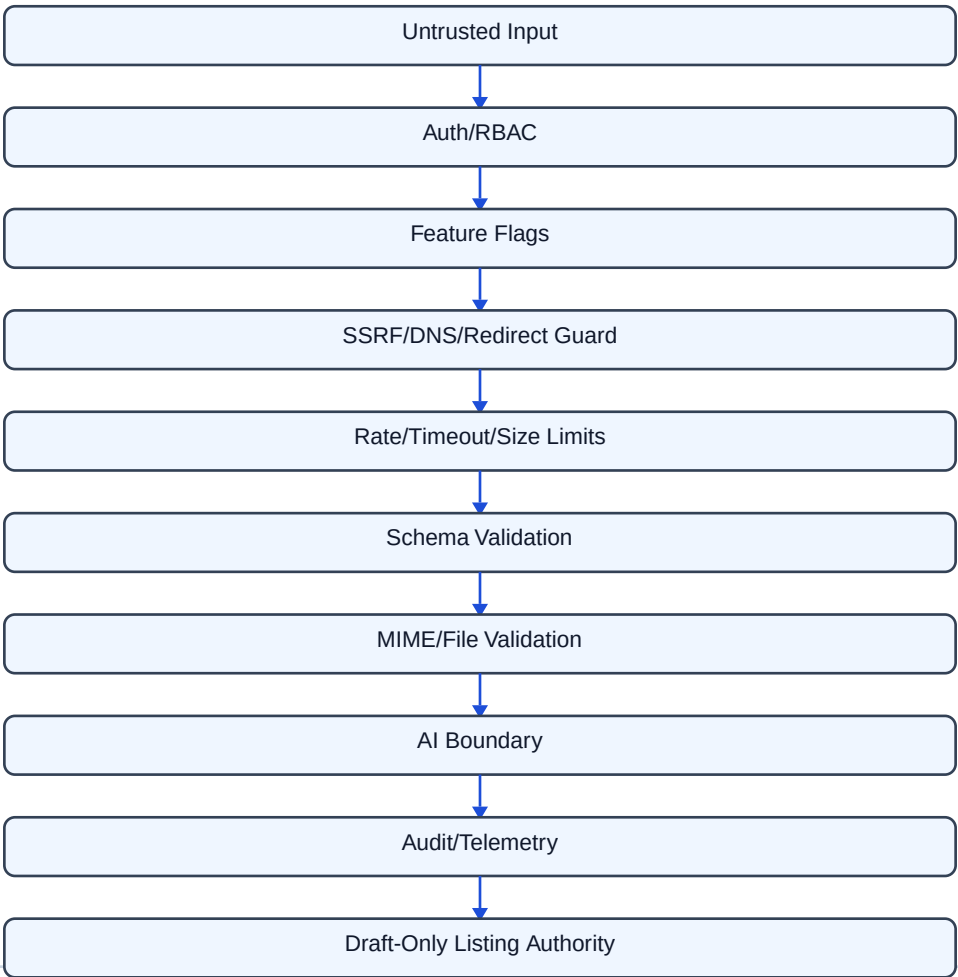
Repeat creation is idempotent. If `created_listing_id` is already present and belongs to the provider, the existing draft id is returned instead of creating another Listing. The resulting `Listing.status` is `Draft` and is not automatically published.

Figure 7. Draft creation sequence.



Chapter 27 - Security Architecture

Figure 8. Security defense-in-depth model.



All external content is treated as untrusted. The secure retrieval layer defaults to HTTPS, rejects embedded credentials, blocks local and cloud metadata hostnames, resolves DNS, rejects unsafe IPv4/IPv6 ranges, pins the request to a prevalidated address, revalidates redirect targets, strips sensitive headers on cross-origin redirects, enforces timeout caps, enforces response-size caps, restricts content types, rate limits retrieval, and writes security audit events on block.

Blocked ranges include unspecified IPv4, private IPv4 10/8, 172.16/12, 192.168/16, carrier-grade NAT 100.64/10, loopback 127/8, link-local 169.254/16, AWS/GCP/Azure/Alibaba metadata addresses captured in traffic, IPv6 loopback, link-local, and unique-local ranges.

File and media controls include MIME and magic-byte validation, size constraints, content hashing, duplicate detection, and storage through RENTipid-managed paths. AI controls include safe context construction, prohibited-field stripping, explicit untrusted-source delimiters, structured output validation, and tool allowlisting. Authorization is re-resolved server-side for corrections, rights confirmation, and draft creation.

Chapter 28 - AI Operating Boundary

Table 19. AI boundary

AI may assist with	AI may not do
Semantic amenity mapping.	Make ownership decisions.
Property category suggestions.	Approve KYC or publication.
Structured fact extraction support.	Override policy, duplicate, or security controls.
Missing-field identification.	Fabricate missing factual data.
Conflict explanations and summaries.	Browse unrestricted external sources.
Original description drafting from verified facts.	Mutate databases directly or create published listings.

Deterministic policy authority remains final. If AI is disabled through configuration, if AI output fails validation, the import continues through deterministic processing, if possible.

Chapter 29 - Media, Location & Duplicate Intelligence

Media validation accepts image/jpeg, image/png, image/webp, and image/gif with a default max size of 10 MB and minimum size of 100 bytes. MIME sniffing checks JPEG, PNG, GIF, and WEBP magic bytes and validates declared MIME compatibility.

Duplicate detection uses same source reference, coordinate proximity under 50 meters, same-provider city with title similarity ≥ 0.7 , and different-provider same-city title similarity ≥ 0.85 . Match levels are exact at scores ≥ 0.95 , likely at ≥ 0.75 , possible at ≥ 0.50 . Exact matches block readiness; likely and possible matches require review.

Chapter 30 - Idempotency & Recovery

Job idempotency is enforced by ListingImportJob.idempotency_key. Media dedupe uses content_sha256 and per-job unique constraints. Draft idempotency uses ListingImportJob.created_listing_id and returns an existing draft if present. Retryable failures increment retry_count and move to FAILED_FINAL when retry_count reaches max_retries, which defaults to 3.

Chapter 31 - API / Server Action / UI Integration

The provider import route is `src/app/dashboard/provider/listings/import/page.tsx`. It renders `src/components/listings/listingbridge/ListingBridgeWizard.tsx` and uses `src/lib/listingbridge/ui/actions.ts` to discover available connectors. Browser interactions call Server Actions in `src/app/dashboard/provider/listings/import/actions.ts`.

Table 20. UI and server responsibilities

Component	Responsibility
ListingBridgeWizard	Source selection, rights confirmation UI, review table, corrections, draft-ready screen, editor link.
startImportAction	Authenticates user, creates durable job, runs test connector path in controlled environments, stores canonical payload and field records.
saveCorrectionAction	Authenticates user, verifies ownership, persists field resolution and verified field state.
confirmRightsAction	Authenticates user, verifies ownership, persists rights confirmation and authorization audit event.
createNativeDraftAction	Authenticates user, verifies ownership and idempotency, delegates to draft creation service.
ListingBridgeDraftCreationService	Rebuilds/evaluates readiness, maps payload, invokes <code>ListingService.createDraft</code> , completes job.

Figure 9. Draft creation sequence.



Chapter 32 - Testing & Acceptance

Controlled OAT defect repair evidence records ListingBridge Test Suite 32/32 suites passing and 212/212 tests passing. Local and Preview acceptance records carry 52/52 mandatory acceptance criteria PASS. Acceptance categories include functional behavior, mapping/data quality, authorization/rights, security, media/location, duplicate/idempotency, AI, UX/resilience, audit, and regression.

Table 21. Acceptance summary matrix

Category	IDs	Count	Status
Confidence & Review	LB-CONF-001 to LB-CONF-008	8	PASS
Core Functionality	LB-FUNC-001 to LB-FUNC-008	8	PASS
Connectors & Registry	LB-CONN-001 to LB-CONN-004	4	PASS
Security & SSRF	LB-SEC-001 to LB-SEC-008	8	PASS
Media Intelligence	LB-MEDIA-001 to LB-MEDIA-004	4	PASS
Location & PSGC	LB-LOC-001 to LB-LOC-004	4	PASS
Duplicate Intelligence	LB-DUP-001 to LB-DUP-004	4	PASS
UX & Provider Actions	LB-UX-001 to LB-UX-004	4	PASS
Unified AI & Fallback	LB-AI-001 to LB-AI-004	4	PASS
Observability & Health	LB-OBS-001 to LB-OBS-004	4	PASS

Chapter 33 - Release Lifecycle G1-G13

Figure 10. Release lifecycle G1-G13.

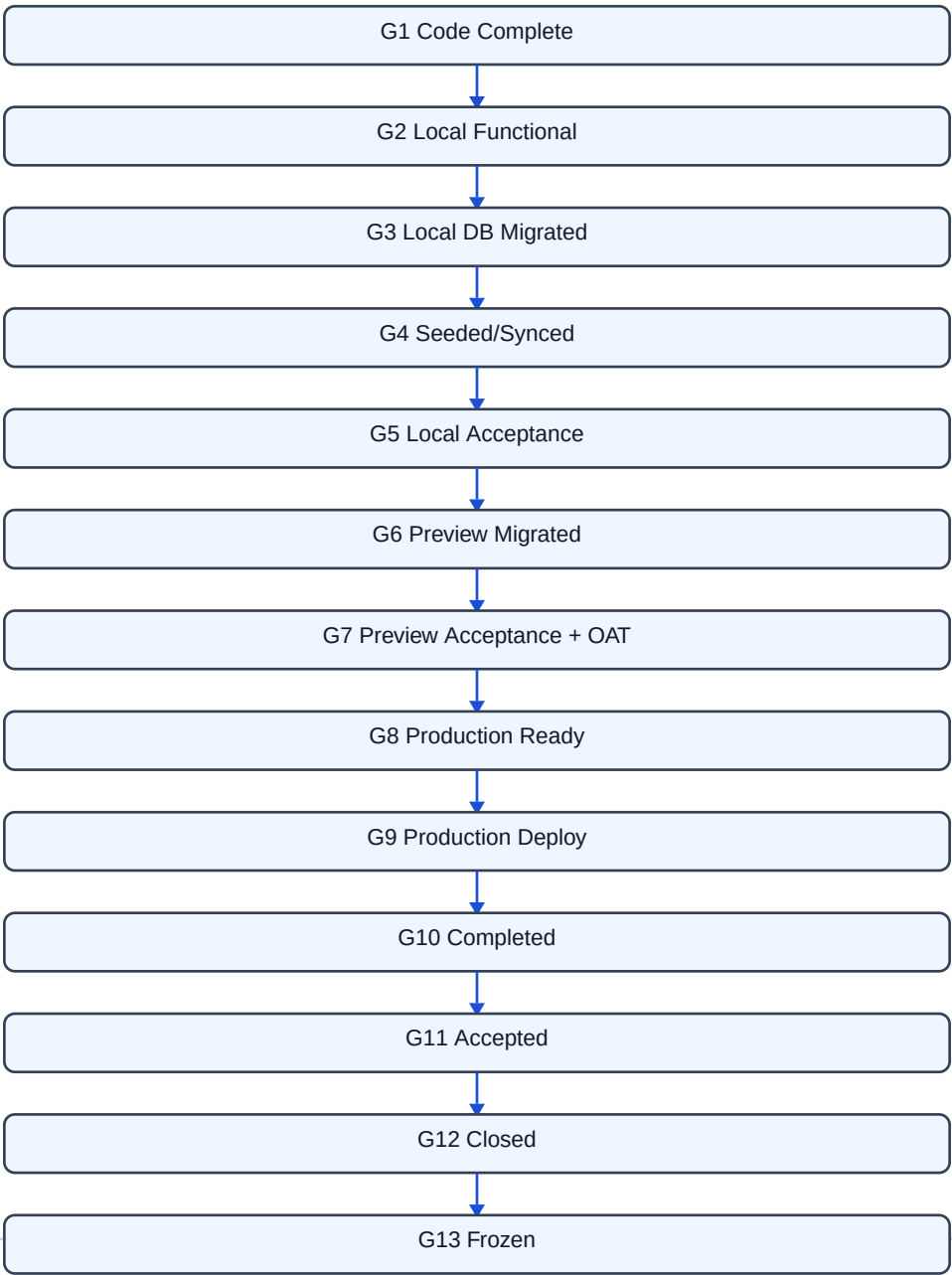


Table 22. Release gate record

Gate	Purpose	Final status	Important evidence
G1 CODE COMPLETE	Final code complete and independently repaired after OAT findings.	PASS	OAT defect repair evidence and 32/32 suites.
G2 LOCAL FUNCTIONAL	Application and import functions operate locally.	PASS	Local functional evidence.
G3 LOCAL DATABASE MIGRATED	Local DB has migration objects.	PASS	Local database migrated evidence.

Controlled Technical & Operational Reference401.0

Gate	Purpose	Final status	Important evidence
G4 LOCAL REQUIRED DATA SEEDED/SYNCD	Feature flags and required data synced.	PASS	Seed/sync evidence.
G5 LOCAL ACCEPTANCE PASS	Mandatory local acceptance verified.	PASS	52/52 acceptance criteria.
G6 PREVIEW MIGRATED	Preview database migration applied.	PASS	Preview migration evidence.
G7 PREVIEW ACCEPTANCE PASS	Automated Preview plus human OAT.	PASS	Preview acceptance and Owner OAT.
G8 PRODUCTION- READY	Release readiness and rollback posture.	PASS	Production readiness evidence.
G9 PRODUCTION DEPLOYMENT/ VERIFICATION	Production deployed and verified.	PASS	Deployment dpl_G4kvoqxoMUUnDMQZFBirfEwt5Ch3.
G10 COMPLETED	Technical completion.	PASS	Completion record.
G11 ACCEPTED	Owner business acceptance.	PASS	Owner acceptance record.
G12 CLOSED	Module closure.	PASS	Closure record.
G13 VERSION FROZEN	Immutable release baseline.	PASS	Freeze record and tag.

IMPORTANT

No-substitution rule: build is not local functional; local functional is not acceptance; Preview deployment is not Preview acceptance; automated Preview is not Owner OAT; production-ready is not production deployment; deployment is not accepted; closed is not frozen.

Chapter 34 - Deployment & Frozen Release Record

Table 23. Frozen release identifiers

Identifier	Value
Application SHA	a8647df71aa9c610027054e2016fd73b53f3b238
Production deployment	dpl_G4kvoqxoMUUnDMQZFBirEwt5Ch3
Production URL	https://www.rentipid.com.ph
Release tag	listingbridge-v1.0.0-frozen
Freeze docs commit	450e74435b95b1d8e7d431f46c3b1466204b4ab4
Canonical migration	20260831000000_add_listingbridge_import_job_foundation
Release state	VERSION FROZEN

The application SHA identifies the deployed and accepted application release. D commits after that point preserve audit history and do not replace the accepted a after freeze requires controlled change and a new lifecycle sequence.

Part V - Business Governance & Future Roadmap

Chapter 35 - Governance & Change Control

ListingBridge v1.0 is frozen. No informal modification to source code, tests, schema, production configuration, feature flags, deployment state, or release evidence is permitted as an in-place change to the baseline.

- Open a controlled change request.
- Assess user, operational, security, database, and business impact.
- Reopen relevant gates and produce new evidence.
- Version the changed module rather than mutating the frozen record.
- Maintain owner acceptance and release-management traceability.

Chapter 36 - Future OTA Connector Strategy

BUSINESS NOTE

Airbnb, Agoda, Booking.com, PMS/channel-manager production connectors, continuous channel synchronization, reservation write-back, and broad price write-back are NOT INCLUDED IN LISTINGBRIDGE v1.0 FROZEN RELEASE.

The preferred future strategy is authorized API/OAuth or approved partner integration -> Connector Adapter -> existing ListingBridge canonical pipeline -> Provider Review -> Native RENTipid Draft. Future work should not bypass access controls, provider authorization, KYC, publication review, SSRF controls, or security/legal review.

Part VI - Reference Appendices

Chapter A - Complete Terminology / Glossary

Table 24. Glossary

ListingBridge	RENTipid subsystem for importing external listing facts into a native draft after review.
Provider	Authenticated RENTipid user or business profile authorized to create listings.
Listing	Native RENTipid marketplace record controlled by existing listing lifecycle.
Draft	Unpublished Listing.status created before submission/review/publication.
Import Job	Durable ListingImportJob record tracking source, status, payload, retry, and draft linkage.
Connector	Adapter implementing ListingBridgeConnector methods for a source mode.
Canonical Import Contract	rentipid.listingbridge.v1 normalized data shape used before draft mapping.
Provenance	Field/source evidence including hashes, timestamps, and authority.
Confidence	Field status used for review and readiness.
VERIFIED	Provider or deterministic trusted confirmation.
HIGH_CONFIDENCE	Strong imported/normalized confidence.
REVIEW_RECOMMENDED	Needs provider inspection.
CONFLICT	Contradictory or invalid data requiring resolution.
MISSING	Expected information not found.
PROHIBITED	Data excluded by policy/security rules.
Authorization	Server-side confirmation of actor rights or connector access.
Rights Confirmation	Provider attestation of property authority, media rights, and accuracy responsibility.
RBAC	Role-based access control.
SSRF	Server-side request forgery; blocked by URL/DNS/IP/redirect policy.
Idempotency	Repeat-safe behavior returning existing job or draft rather than duplicating.
Duplicate Detection	Signals that identify same or similar existing listings/imports.
Feature Flag	SystemSetting runtime switch controlling ListingBridge capabilities.
Kill Switch	LISTINGBRIDGE_GLOBAL=false production-safe disable state.
OAT	Owner Operational Acceptance Test.
Preview	Non-production Vercel deployment/database used for acceptance.

ListingBridge	RENTipid subsystem for importing external listing facts into a native draft after review.
Production	Live rentipid.com.ph deployment and database.
Migration	Prisma schema change applied through migration history.
Baseline	Controlled known-good state before applying release migration.
Freeze	Immutable accepted version state requiring controlled change for modification.
Controlled Change	Formal post-freeze change process with impact assessment and evidence.
AI Mapping	Bounded semantic assistance for taxonomy and review support.
Draft-Only Boundary	Rule that ListingBridge creates only Draft listings.
PMS	Property management system or channel manager source.
OTA	Online travel agency such as Airbnb, Agoda, or Booking.com.
API	Programmatic source interface.
OAuth	Authorization flow for delegated API access.
MIME	Media/content type identifier verified against bytes.
PSGC	Philippine Standard Geographic Code hierarchy.
Audit Event	Durable evidence of lifecycle, security, correction, or draft action.
Trace ID	Correlation identifier for logs, metrics, and support investigation.
ListingService	Native RENTipid listing authority containing createDraft lifecycle.

Chapter B - Feature Flag Reference

Table 25. Complete feature flag reference

Key	Purpose	Frozen Production Value	Operational Effect	Dependencies	Change-Control Note
LISTINGBRIDGE_GLOBAL	Master ListingBridge kill switch and rollout gate.	false	No ListingBridge connectors are offered; manual listing creation remains available. / ListingBridge import entry points and connector evaluation are permitted subject to child flags.	All ListingBridge capabilities.	Frozen release flag. Change only through controlled rollout.
LISTINGBRIDGE_FILE_IMPORT	Enables Tier 3 structured file import capability.	true	Structured file import is unavailable. / Structured file connectors can be evaluated when global flag also permits.	LISTINGBRIDGE_GLOBAL.	Do not confuse enabled capability with production-wide public rollout.
LISTINGBRIDGE_URL_IMPORT	Enables Tier 4 public URL retrieval.	false	URL import is unavailable. / SSRF-guarded URL retrieval can run for approved connectors.	LISTINGBRIDGE_GLOBAL, retrieval policy, SSRF controls.	High caution due to external network exposure.
LISTINGBRIDGE_API_CONNECTORS	Enables API or OAuth partner connectors.	false	OAuth/API connector paths are unavailable. / Approved server-side API connectors may be evaluated.	LISTINGBRIDGE_GLOBAL, connector credentials, compliance approval.	No live Airbnb, Agoda, or Booking.com connector exists in v1.0.
LISTINGBRIDGE_MEDIA_IMPORT	Enables media retrieval, validation, hashing, and storage.	true	Imports can continue as text/detail-only where supported. / External media candidates may be ingested after validation.	Storage provider, media MIME validation, source retrieval.	Disable if storage or MIME validation anomaly is observed.
LISTINGBRIDGE_AI_MAPPING	Enables bounded semantic mapping assistance.	true	Deterministic mapping and provider review continue without AI. / AI may suggest amenities, categories, summaries, and original descriptions from verified facts.	Unified AI adapter and read-only/draft-only tools.	AI output is advisory and never policy-authoritative.
LISTINGBRIDGE_AVAILABILITY_IMPORT	Enables external availability/calendar import.	false	Availability data is not imported from external calendars. / Approved availability connector capability may be used.	LISTINGBRIDGE_GLOBAL and connector availability support.	Reservation write-back and continuous sync are not v1.0 scope.

Chapter C - Import Job State Reference

Table 26. Complete import job state transition table

State	Meaning	Expected transition	Operator concern	Recovery / escalation notes
CREATED	Durable job record exists.	AUTHORIZING or FETCHING.	Job should not remain idle for long.	Retry or cancel only through controlled support flow.
AUTHORIZING	Provider rights or connector authorization is being verified.	FETCHING.	Repeated authorization failures may indicate user action or connector issue.	Ask provider to retry authorization; do not bypass rights confirmation.
FETCHING	Connector or secure retrieval is obtaining source payload.	EXTRACTING.	Timeouts and connector errors are expected transient concerns.	Retry if retryable; security failures become final.
EXTRACTING	Structured facts are being read from payload.	NORMALIZING.	Malformed payloads can produce missing fields.	Escalate only if repeated for known-good source.
NORMALIZING	Facts are mapped into canonical contract.	PROCESSING_MEDIA or VALIDATING.	Unexpected confidence spikes or prohibited fields require review.	Inspect normalized field records and rejected fields.
PROCESSING_MEDIA	Media candidates are downloaded, validated, deduped, and stored.	VALIDATING, NEEDS_REVIEW, or READY_FOR_DRAFT.	Storage, MIME, or partial failures.	Text import may survive partial media failure; missing required photo can block readiness.
VALIDATING	Readiness, policy, location, duplicate, and field checks run.	NEEDS_REVIEW or READY_FOR_DRAFT.	Blocking fields must be resolved before draft.	Use field/resolution records to guide provider.
NEEDS_REVIEW	Provider must inspect, correct, or confirm fields.	READY_FOR_DRAFT or VALIDATING.	Support should not edit provider assertions casually.	Provider correction is preferred.
READY_FOR_DRAFT	Server-side readiness has no blockers.	CREATING_DRAFT.	If job regresses, review duplicate/location/media blockers.	Proceed to draft only after rights confirmation.
CREATING_DRAFT	Draft creation through native Listing authority is underway.	COMPLETED or failure state.	Stuck jobs may have native draft already created.	Check created_listing_id before any retry.
COMPLETED	Native RENTipid draft has been created and linked.	Terminal.	No further import mutation expected.	Provider continues normal listing editing workflow.
FAILED_RETRYABLE	Transient failure occurred and may be retried.	Prior processing state, FAILED_FINAL, or CANCELLED.	Retry count and next_attempt_at matter.	Retry is bounded by max_retries=3.
FAILED_FINAL	Non-retryable or exhausted failure.	Terminal.	Security, corrupt data, or repeated retry exhaustion.	Manual listing fallback; preserve audit evidence.
CANCELLED	Provider or operator cancelled import.	Terminal.	No draft should be created from a cancelled job.	Manual fallback remains available.

Chapter D - Acceptance Test Catalogue

Table 27. All 52 frozen mandatory acceptance IDs

Acceptance ID	Scenario	Status
LB-CONF-001	Confidence states exist and are rendered consistently.	PASS
LB-CONF-002	Verified/high-confidence fields remain non-blocking.	PASS
LB-CONF-003	Review-recommended fields produce warnings.	PASS
LB-CONF-004	Conflicts are visible and actionable.	PASS
LB-CONF-005	Missing required fields block readiness.	PASS
LB-CONF-006	Prohibited fields are excluded from active listing content.	PASS
LB-CONF-007	Provider corrections update confidence to verified.	PASS
LB-CONF-008	Review snapshots preserve field provenance.	PASS
LB-FUNC-001	Import job creation persists durable state.	PASS
LB-FUNC-002	Manual listing creation remains independent.	PASS
LB-FUNC-003	Canonical contract validation succeeds for valid payloads.	PASS
LB-FUNC-004	Extraction and normalization produce mapped fields.	PASS
LB-FUNC-005	Provider rights confirmation is required.	PASS
LB-FUNC-006	Native draft creation uses ListingService authority.	PASS
LB-FUNC-007	Draft status remains Draft only.	PASS
LB-FUNC-008	Import flow can resume from durable state.	PASS
LB-CONN-001	Connector descriptors validate contract shape.	PASS
LB-CONN-002	Environment resolution distinguishes Preview from Production.	PASS
LB-CONN-003	Internal test connector is available for controlled Preview testing.	PASS
LB-CONN-004	Internal test connector is disabled in Production.	PASS
LB-SEC-001	HTTPS-only default retrieval policy enforced.	PASS
LB-SEC-002	Loopback/private/link-local addresses are blocked.	PASS
LB-SEC-003	Cloud metadata endpoints are blocked.	PASS
LB-SEC-004	DNS failure fails closed.	PASS

Acceptance ID	Scenario	Status
LB-SEC-005	Redirect targets are revalidated.	PASS
LB-SEC-006	Sensitive headers are stripped/redacted.	PASS
LB-SEC-007	Response size, timeout, and content-type limits are enforced.	PASS
LB-SEC-008	Ownership mismatch blocks cross-tenant access.	PASS
LB-MEDIA-001	Valid JPEG/PNG/WEBP/GIF media is accepted.	PASS
LB-MEDIA-002	Invalid MIME and executable-like content is rejected.	PASS
LB-MEDIA-003	SHA-256 deduplication prevents duplicate media writes.	PASS
LB-MEDIA-004	Partial media failures do not erase valid import details.	PASS
LB-LOC-001	Philippine address normalization succeeds.	PASS
LB-LOC-002	Coordinates outside Philippine bounds conflict for PH listings.	PASS
LB-LOC-003	Missing city/address blocks readiness.	PASS
LB-LOC-004	Address review state is surfaced to provider.	PASS
LB-DUP-001	Previously imported same source is exact match.	PASS
LB-DUP-002	Nearby same-provider coordinates are likely/exact duplicate signals.	PASS
LB-DUP-003	Similar title and city produce possible-match review.	PASS
LB-DUP-004	Unrelated listings return no match.	PASS
LB-UX-001	Provider dashboard links to create and import listing.	PASS
LB-UX-002	Source selector and authorization step operate.	PASS
LB-UX-003	Provider correction UI saves edited fields.	PASS
LB-UX-004	Manual fallback link opens the native listing wizard.	PASS
LB-AI-001	AI cannot fabricate missing required facts.	PASS
LB-AI-002	Invalid AI structured output fails safely.	PASS
LB-AI-003	AI cannot publish, approve, or bypass policy.	PASS
LB-AI-004	Core import works with AI disabled.	PASS
LB-OBS-001	Metrics use bounded safe dimensions.	PASS
LB-OBS-002	Health report summarizes flags, connector health, and import totals.	PASS

Acceptance ID	Scenario	Status
LB-OBS-003	Alerts trigger for SSRF, draft failure, and connector failure thresholds.	PASS
LB-OBS-004	Structured logging and redaction avoid secrets and PII dimensions.	PASS

Chapter E - Support Troubleshooting Matrix

Table 28. Detailed operator support table

Issue	Diagnostic evidence	Operator response	Escalation
Connector outage	connector failure metrics, health state DEGRADED/UNHEALTHY.	Disable affected connector capability if necessary; preserve manual fallback.	Engineering/SRE and partner owner.
Repeated import failure	FAILED_RETRYABLE trend, retry_count near max_retries.	Classify error category; verify source is allowed and reachable.	Engineering if deterministic reproducible failure.
SSRF/security block spike	listingbridge_ssrf_block_total >= 10 or SECURITY_BLOCKED audit events.	Investigate actor/source patterns; rate limit or suspend abusive account.	Security/SOC immediately.
Worker interruption	Stale FETCHING/ PROCESSING_MEDIA/ CREATING_DRAFT with lock timestamps.	Resume from durable state; check created_listing_id before retry.	Engineering if state transition invalid.
Media storage failure	MEDIA_PROCESSING_EXCEPTION, storage errors, media_failure metric.	Temporarily disable media import if systemic.	Storage/SRE.
Duplicate anomaly	Unexpected EXACT_MATCH or widespread false positives.	Review signals: source hash, coordinate proximity, title/city similarity.	Product/Engineering for threshold tuning.
Draft creation failure	DRAFT_READINESS_FAILED or LISTING_AUTHORITY_CREATION_FAILED.	Verify required fields, Category resolution, ListingService health.	Engineering if native service rejects valid payload.
Database issue	/api/health database not connected, Prisma errors.	Follow production DB incident process; avoid destructive rollback.	SRE/DBA.
Runtime regression	Vercel runtime errors after deployment.	Use global kill switch; consider Vercel rollback.	Release manager.

Chapter F - Security Control Matrix

Table 29. Security control matrix

Threat	Control	Implementation component	Detection	Operator response
SSRF to internal network	Protocol, DNS, IP range, redirect, and pinned-address validation.	SsrfProtectionService; SecureHttpRetrievalEngine.	SECURITY_BLOCKED audit; ssrf block metric.	Treat spikes as security incident.
Credential leakage	Server-only credentials; sensitive header stripping and log redaction.	credential-boundary; secure retrieval; structured logger.	Redaction tests and audit review.	Rotate affected secret if exposure suspected.
Prompt injection	External text delimited as untrusted and AI tools restricted.	SafeAiContextBuilder; AiToolGateway tools.	AI fallback/validation failures.	Inspect source payload and disable AI mapping if needed.
Malicious media/file	MIME sniffing, size limits, content hashing.	MediaSecurityValidator; file-preflight.	media_failure metric and rejected asset rows.	Block source or guide provider to manual upload.
Cross-tenant access	Server session re-resolution and provider_id ownership check.	Server actions; draft creation service.	OWNERSHIP_MISMATCH errors.	Security escalation if malicious.
Duplicate draft creation	created_listing_id and idempotency guard.	ListingImportJob unique constraints; draft service.	Duplicate exact match signals.	Return existing draft; do not create new one manually.

Chapter G - File / Component Reference

Table 30. Developer file and component quick reference

Canonical contract	Defines rentipid.listingbridge.v1 schema, confidence states, source identity, media, provenance.	src/lib/listingbridge/types/canonical-contract.ts
Job state types	Lists job/asset/resolution/audit states and legal transitions.	src/lib/listingbridge/types/job-state.ts
Connector interface	Defines ListingBridgeConnector method contract.	src/lib/listingbridge/connectors/types.ts
Connector descriptor	Defines capability, environment, compliance, health, timeout, retry, and feature controls.	src/lib/listingbridge/connectors/descriptor.ts
Connector registry	Registers, filters, and evaluates connector availability.	src/lib/listingbridge/connectors/registry.ts
Feature flags	Maps capabilities to SystemSetting flags and manual-flow independence.	src/lib/listingbridge/connectors/feature-flags.ts
Environment resolver	Distinguishes VERCEL_ENV Preview/Production and fails closed.	src/lib/listingbridge/connectors/environment.ts
Internal test connector	Deterministic fixture connector disabled in Production.	src/lib/listingbridge/connectors/test-connector.ts
Seed settings	Idempotently seeds required ListingBridge SystemSetting rows.	src/lib/listingbridge/connectors/seed.ts
SSRF service	Validates URLs, hostnames, DNS, IP ranges, and redirects.	src/lib/listingbridge/security/ssrf-protection.ts
Secure retrieval	Pinned-address HTTP retrieval with header stripping, content limits, audit, and rate limiting.	src/lib/listingbridge/retrieval/secure-http-retrieval.ts
Retrieval policy	Default and hard retrieval limits.	src/lib/listingbridge/retrieval/policy.ts
Rate control	Database-backed retrieval rate limit helper.	src/lib/listingbridge/retrieval/rate-control.ts
Extraction	Alias-based structured fact extraction.	src/lib/listingbridge/extraction/structured-extractor.ts
Normalization pipeline	Maps facts, filters prohibited data, detects conflicts, optional AI assistance.	src/lib/listingbridge/normalization/pipeline.ts
Review snapshot	Builds provider review model and readiness summary.	src/lib/listingbridge/review/review-snapshot-engine.ts
Readiness engine	Evaluates rights, duplicate, location, media, status, and field blockers.	src/lib/listingbridge/review/draft-readiness-engine.ts
Provider corrections	Applies validated provider corrections.	src/lib/listingbridge/review/provider-correction-service.ts
Import repository	Durable job/source/field/status persistence.	src/lib/listingbridge/repository/listing-import-repository.ts
Draft service	Server-side draft creation and idempotency.	src/lib/listingbridge/draft/draft-creation-service.ts
Draft mapper	Maps review snapshot to native draft payload.	src/lib/listingbridge/draft/draft-payload-mapper.ts
Media security	MIME sniffing, size limits, and hash generation.	src/lib/listingbridge/media/media-security.ts
Media pipeline	Media candidate retrieval, validation, storage, and duplicate handling.	src/lib/listingbridge/media/media-ingestion-pipeline.ts
Location intelligence	Address normalization and Philippine bounds checks.	src/lib/listingbridge/location/location-intelligence.ts

Canonical contract	Defines rentipid.listingbridge.v1 schema, confidence states, source identity, media, provenance.	src/lib/listingbridge/types/canonical-contract.ts
Duplicate detector	Exact/likely/possible duplicate signal logic.	src/lib/listingbridge/duplicates/duplicate-detector.ts
AI tools	Read-only and draft-only ListingBridge AI tools.	src/lib/listingbridge/ai/tools.ts
Safe AI context	Filters prohibited fields and delimits untrusted source data.	src/lib/listingbridge/ai/safe-context-builder.ts
AI service	Review summary, missing fields, conflict explanation, mapping, description draft.	src/lib/listingbridge/ai/listingbridge-ai-service.ts
Metrics	Bounded-dimension metrics collector.	src/lib/listingbridge/observability/metrics.ts
Health	ListingBridge health diagnostics service.	src/lib/listingbridge/observability/health.ts
Alerts	SSRF, draft failure, and connector outage alert conditions.	src/lib/listingbridge/observability/alerts.ts
Provider import route	Renders ListingBridge import page.	src/app/dashboard/provider/listings/import/page.tsx
Provider server actions	Start import, save correction, confirm rights, create native draft.	src/app/dashboard/provider/listings/import/actions.ts
Wizard UI	Client source selection, review, correction, draft handoff.	src/components/listings/listingbridge/ListingBridgeWizard.tsx
Prisma schema	ListingBridge models and enums.	prisma/schema.prisma
Migration SQL	Canonical additive ListingBridge migration.	prisma/migrations/20260831000000_add_listing_bridge_import_job_foundation/migration.sql

Chapter H - Database Entity Reference

Table 31. Database entity reference

Entity	Purpose	Key relationships	Lifecycle role	Important constraints
ListingImportJob	Durable root import job and status state.	provider_id -> User; created_listing_id -> Listing.	Owns canonical payload, retry metadata, status, and draft linkage.	Unique idempotency_key and created_listing_id; indexes by provider/status/source.
ListingImportSource	Per-source retrieval/provenance record.	job_id -> ListingImportJob.	Records connector, tier, source mode, version, hashes, and retrieval metadata.	Indexes source reference and identifier.
ListingImportField	Field-level normalized value, confidence, and validation record.	job_id -> ListingImportJob; optional source_id -> ListingImportSource.	Supports provider review, readiness, and provenance.	Unique job_id+field_name; indexes confidence and blockers.
ListingImportAsset	Media candidate, validation, storage, and hash record.	job_id -> ListingImportJob.	Tracks media success, rejection, duplicates, and storage path.	Unique job_id+source_reference_hash and job_id+content_sha256.
ListingImportResolution	Provider/system resolution of a field.	job_id -> ListingImportJob; resolved_by_user_id -> User.	Persists corrections and rights confirmation.	Unique job_id+field_name; indexed by resolver/time.
ListingImportAuditEvent	Append-only import audit trail.	job_id -> ListingImportJob; actor_user_id -> User; audit_log_id -> AuditLog.	Records lifecycle, authorization, security, correction, and draft events.	Indexed by job/time, actor/time, and central audit link.

Chapter I - Release / Evidence Reference

Table 32. Release and evidence reference

Item	Value
Application SHA	a8647df71aa9c610027054e2016fd73b53f3b238
Freeze docs commit	450e74435b95b1d8e7d431f46c3b1466204b4ab4
Release tag	listingbridge-v1.0.0-frozen
Preview deployment	dpl_37GAd7SmgPunNayBcsvxuR6214kt
Production deployment	dpl_G4kvoqxoMUUnDMQZFBirEwt5Ch3
Production URL	https://www.rentipid.com.ph
Canonical migration	20260831000000_add_listingbridge_import_job_foundation
Local acceptance	52/52 PASS
Preview automated acceptance	52/52 PASS
Owner OAT	PASS
Owner business acceptance	PASS
Evidence root	docs/listingbridge/evidence/ListingBridge-v1.0/
Evidence directories	00-baseline through 15-freeze, including architecture, code complete, local/preview acceptance, OAT, production readiness, deployment, completion, acceptance, closure, and freeze.

Chapter J - Quick Reference Cards

Table 33. Provider user quick reference

Do	Remember
Use Import Existing Listing when available.	You must confirm authority and media rights.
Review every imported field.	Please Review means inspect before relying on it.
Correct missing or conflicting fields.	ListingBridge creates only Draft listings.
Open the draft in Listing Editor.	Submit for review separately when ready.

Table 34. Support operator quick reference

Check	Action
Feature flags	GLOBAL=false disables import safely.
Job status	Use status, retry_count, last_error_code, and audit events.
Security blocks	Do not bypass; escalate SSRF spikes.
Draft issues	Check created_listing_id before retry.

Table 35. Developer quick reference

Area	Primary path
Types/contracts	src/lib/listingbridge/types/
Connectors	src/lib/listingbridge/connectors/
Repository	src/lib/listingbridge/repository/listing-import-repository.ts
Draft creation	src/lib/listingbridge/draft/draft-creation-service.ts
UI	src/components/listings/listingbridge/ListingBridgeWizard.tsx

Table 36. Business owner quick reference

Decision point	v1.0 answer
Is it live OTA sync?	No. One-time controlled import to Draft.
Can it publish?	No. Existing approval lifecycle remains authoritative.
Can it be disabled?	Yes. GLOBAL=false.
Are future connectors possible?	Yes, through controlled change and approved integrations.

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